

S1 NL Input

"I struggle with my electricity bill"

user answer

S5 Follow-up Q

"Do you live alone?"

S2 NL → Tag

"60s" → age 60–69
"elderly" → senior

5-Stage Closed Loop

S4 Recommendation

9-dim persona →
GT-1 $\cap$ GT-2, top-K

S3 Query Rewrite

[seniors 60+]
[heating subsidy]

